

Lecture 20: Making Statistical Claims

*Instructor: Jonathan Pipping-Gamón**Authors: JP, RB*

20.1 From Statistical Models to Applications

Most of the course has focused on building statistical models. We have estimated expected points, win probabilities, player abilities, team strengths, treatment effects, and player impact coefficients. Once those models are trained, one step remains: deciding what conclusions the findings justify.

That step is easy to skip. A model output can quickly turn into a claim like

- “The model says go for it on 4th down.”
- “This player is the eighth best player in the league.”
- “The timeout caused the comeback.”
- “This batting average proves the player improved.”
- “The neural network found what coaches should call.”

Each of these takeaways may come from disciplined analysis, but they may also overstate their conclusions by hiding the target, the uncertainty, and the assumptions that produced the estimate.

Claims should match the target, the model, and the uncertainty. This lecture focuses on two common places where sports models get overstated: fourth down recommendations and rankings built from estimated metrics.

20.2 Case Study 1: Fourth Down Decisions

Consider the fourth down decision problem. On fourth down, the offense can keep its offense on the field, attempt a field goal, or punt:

$$d \in \{\text{go, field goal, punt}\}.$$

Going for it can extend the drive, but a failed attempt gives the opponent the ball near the current spot. A field goal attempt can produce three points, but a miss gives up field position. A punt gives up the ball on purpose in exchange for forcing the opponent to start farther away from the end zone.

A model cannot recommend an action until we say what the action should optimize. A natural target is win probability. If x is the current game state, including score, time, field position, yards to go, and team information, the three decision values are

$$WP_{\text{go}}(x), \quad WP_{\text{FG}}(x), \quad WP_{\text{punt}}(x).$$

The counterfactual meaning of “would” matters here. In the actual game, teams choose one action. The other two outcomes are counterfactual. The model estimates what the team’s win probability would be after each possible action:

$$\widehat{WP}_{\text{go}}(x), \quad \widehat{WP}_{\text{FG}}(x), \quad \widehat{WP}_{\text{punt}}(x).$$

Under a win probability objective, the plug in rule is

$$\hat{d}(x) = \arg \max_{d \in \{\text{go}, \text{FG}, \text{punt}\}} \widehat{WP}_d(x).$$

The estimated edge for the selected action is

$$\hat{\Delta}(x) = \widehat{WP}_{\hat{d}}(x) - \max_{d \neq \hat{d}} \widehat{WP}_d(x).$$

This separates two things that are often mixed together. The model estimates decision values. The objective turns those values into a recommendation. That recommendation optimizes estimated win probability under the fitted model, not job security, injury risk, locker room trust, or future play calling value.

Tied, 4th & 3, 35 yards from opponent endzone								
Qtr 1, 10:28 Timeouts: Off 3, Def 3 Point Spread: -2.5, OQ-DQ: -0.49, KQ: 0, PQ: 0								
decision	WP	90% WP gain CI	boot %	success prob	WP if fail	WP if succeed	SD of WP	baseline coach %
Go for it	55.9%	[-1%, 5%]	89%	48%	47.9%	64.8%	8.4%	50.9%
Field goal	54.1%		11%	61%	47.0%	58.5%	5.6%	45.5%
Punt	52.4%		0%					3.6%

Figure 20.1: Example output from Ryan Brill’s fourth down decision model for the Eagles’ fourth-and-3 against the 49ers in the 2022 NFC Championship Game. The table compares the model’s estimated win probability for each action.

20.2.1 Stability of the Recommendation

A fourth down recommendation is typically built from several estimated components:

- the probability of converting, estimated from previous fourth down attempts;
- the probability of making a field goal, based on distance and kicking context;
- the value of a punt, based on field position and subsequent scoring outcomes;
- win probability, based on score, time remaining, field position, team strength, and remaining possessions.

Each component is estimated with uncertainty. In addition, the available data do not capture every relevant feature of the football environment. Factors such as the exact play call, injuries, fatigue, weather, personnel packages, and private information available to the coaching staff may all affect the true value of a decision but remain outside the model.

The point estimates describe the size of the estimated advantage. For example, if going for it has an estimated win probability of 0.559 and kicking a field goal has an estimated win probability of 0.541, then the estimated edge from going for it is 0.018. However, the size of this difference is not, by itself, a measure of the recommendation’s stability. To evaluate stability, we also need to estimate how much the fitted action values vary under changes to the fitted model.

A natural approach is to refit the full modeling pipeline on bootstrap samples. For each bootstrap refit, we recompute the win probability values of the available actions for the same game state and record which action has the highest estimated value. If going for it is selected in nearly every refit, then the recommendation has strong stability. If going for it and kicking a field goal frequently switch places, then the recommendation is less stable, even if the original point estimate favors going for it.

With B bootstrap refits, one useful summary is

$$\hat{\mathbb{P}}(\text{go is best}) = \frac{1}{B} \sum_{b=1}^B \mathbb{1} \left\{ \widehat{WP}_{\text{go}}^{*(b)}(x) > \max(\widehat{WP}_{\text{FG}}^{*(b)}(x), \widehat{WP}_{\text{punt}}^{*(b)}(x)) \right\}.$$

This quantity is the proportion of bootstrap refits in which going for it is estimated to be the best action. Values close to 1 indicate stable support for going for it, while smaller values indicate that alternative actions are often preferred after refitting. The same bootstrap refits can also be used to examine the variability of the estimated action gap. This does not replace the original point estimate; rather, it measures how sensitive the estimated edge is to perturbations of the data. For football play-by-play data, the choice of resampling unit is important. Resampling individual plays may understate uncertainty because plays from the same game can share unobserved context, so resampling drives or full games is preferred.

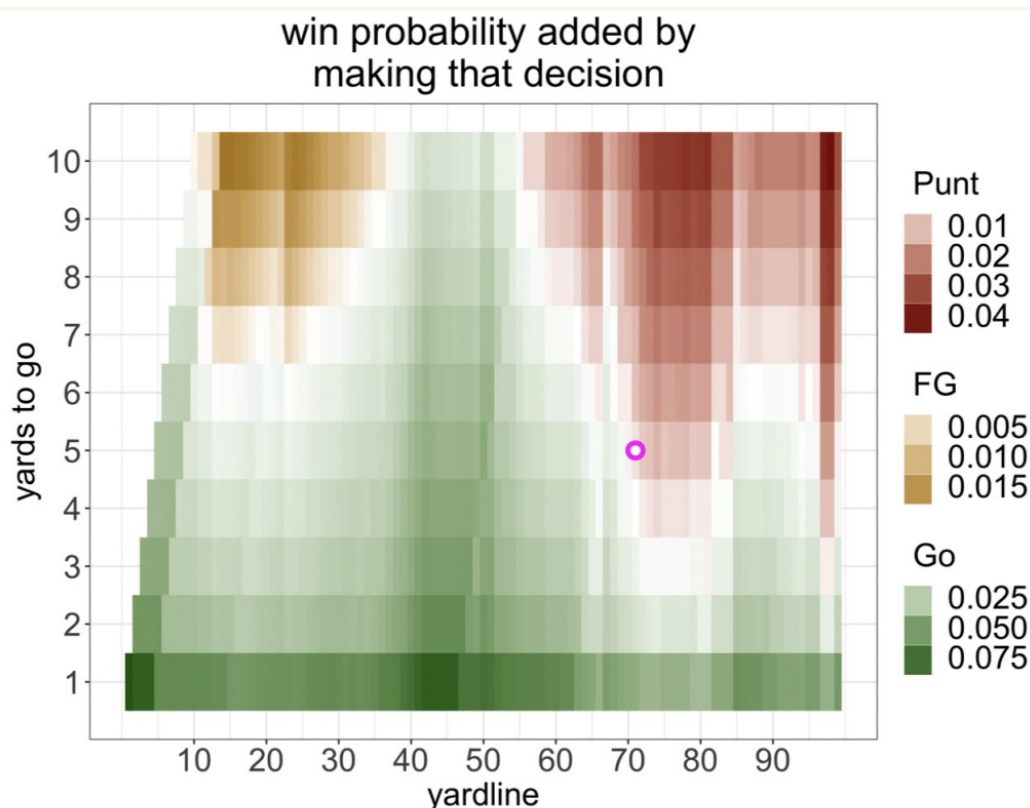


Figure 20.2: Estimated win probability gain from Brill's fourth down decision model. The point estimate gives the magnitude of the edge, while refit variability assesses the stability of that edge.

20.2.2 Reporting Fourth Down Recommendations

The language used to report a fourth down recommendation should reflect both the estimated edge and the stability of that recommendation under refitting. Table 20.1 gives one possible vocabulary.

Label	Pattern	Suggested wording
Clear	Positive edge; stable across refits	The fitted win probability model favors going for it.
Lean	Positive edge; mixed refit support	The point estimate favors going for it, but the call is sensitive.
Too close	Small edge or unstable best action	The model treats the options as essentially tied.
Not covered	Missing context or outside training support	This case is outside the model's checked support.

Table 20.1: Language for fourth down recommendations.

For fourth down decisions, reporting should identify the objective, the magnitude of the estimated edge, and the stability of the recommendation under refitting. The final column is therefore not merely an after-the-fact qualification; it is part of the analysis itself. Rankings raise a similar issue in a different form. An ordered list can appear more certain, and more broadly applicable, than the fitted model actually supports.

20.3 Case Study 2: Rankings From Derived Metrics

In Lecture 14, a Bayesian power rating model was used to estimate team strengths, measured as expected point differential relative to league average. The model did not simply produce a single ordered list. Instead, posterior draws represented plausible values of team strength after observing the data. Within any one posterior draw, teams can be ranked. Across draws, however, the ordering can change. The reporting problem arises when this uncertainty is compressed into a single leaderboard.

If θ_j denotes team j 's strength, then a point ranking orders the estimates $\hat{\theta}_j$. But θ_j is not a generic measure of team quality. In Lecture 14, it was a latent score margin effect from a model with team indicators, home field advantage, and partial pooling. If we change the response, predictors, season window, or prior, the ranking can change because both the definition of strength and the evidence used to estimate it have changed. A posterior ranking can instead report

$$\mathbb{P}(\text{rank}_j \leq 5 \mid \text{data}) \approx \frac{1}{M} \sum_{m=1}^M \mathbb{1}\{\text{rank}(\theta_j^{(m)}) \leq 5\}.$$

A pairwise comparison can report

$$\mathbb{P}(\theta_A > \theta_B \mid \text{data}).$$

These are different claims:

- Team A has the third highest posterior mean.
- Team A is top five in 82% of posterior draws.
- Team A is stronger than Team B in 63% of posterior draws.

The first sentence reports an ordered point estimate. The second reports uncertainty about rank. The third reports pairwise uncertainty. These are different summaries of the same posterior distribution, and they should not be confused with a general ranking of team quality.

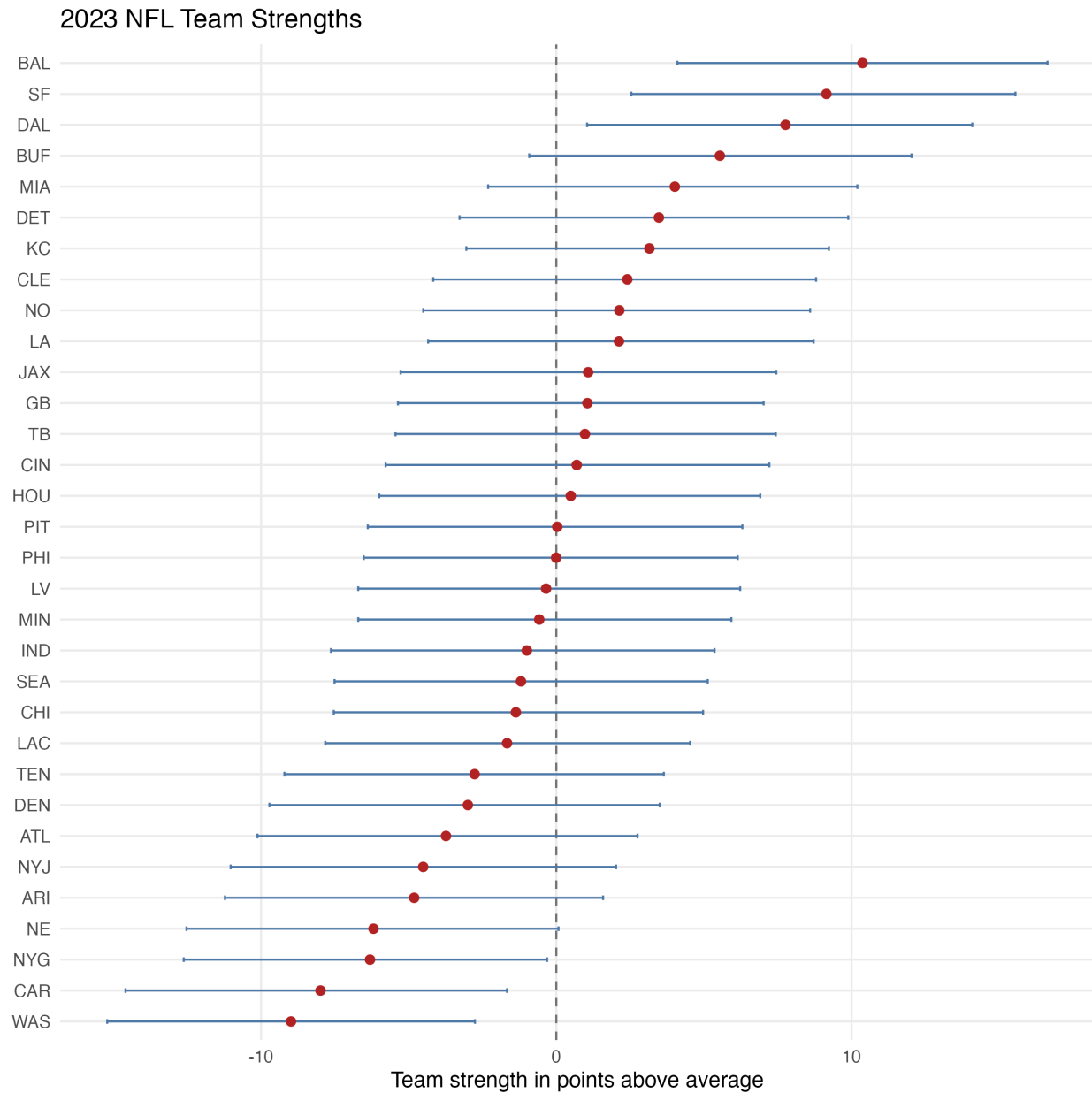


Figure 20.3: Posterior means and 95% credible intervals for team strength from Lecture 14. Exact rankings are less stable when intervals overlap.

20.3.1 Leaderboards Select Extremes

Uncertainty is only part of the issue. A leaderboard also depends on the quantity the model was designed to estimate. Large observed or fitted values often combine signal with favorable noise. This is why shrinkage appeared in several parts of the course:

- empirical Bayes moved noisy batting averages toward a league-learned prior;
- shrinkage estimation reduced joint prediction risk;
- ridge regression stabilized adjusted plus-minus estimates;
- hierarchical Bayesian models pooled related team effects.

Player impact metrics make this issue concrete. Raw plus-minus, adjusted plus-minus, and regularized adjusted plus-minus are different ways to turn on-court scoring data into a player ranking. In RAPM, the response is possession margin, the predictors are lineup indicators, and regularization penalizes large coefficients. The resulting estimate concerns lineup-adjusted point impact in that sample, not total player value.

Three Plus-Minus Estimates

Same 2023-24 possession data; estimates shown in points per 100 possessions

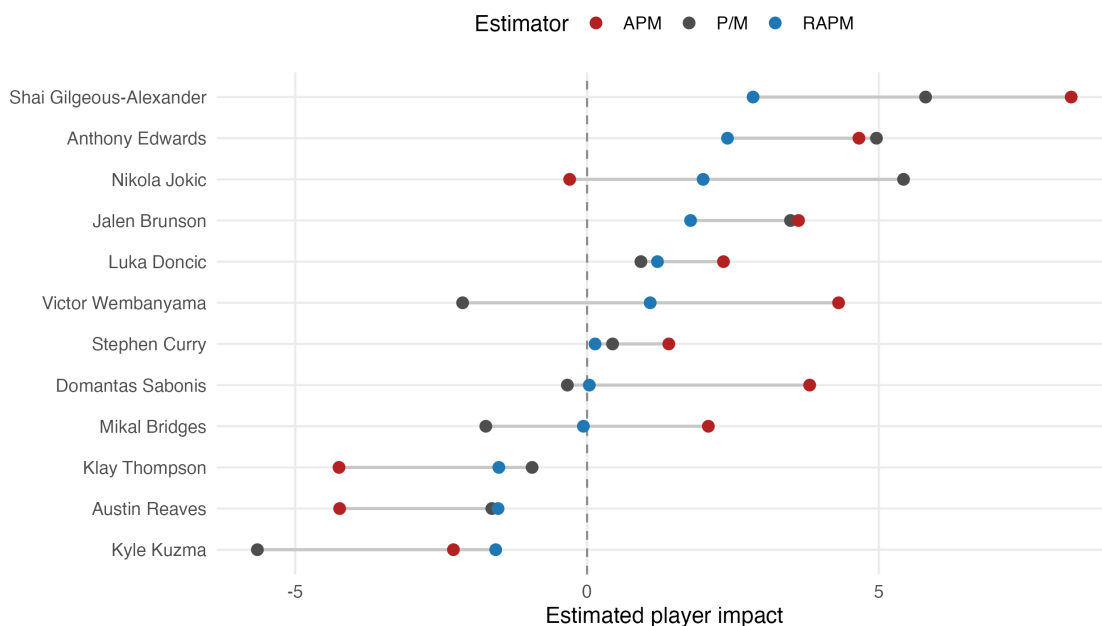


Figure 20.4: Raw plus-minus, adjusted plus-minus, and regularized adjusted plus-minus from Lecture 15. Regularization changes the objective by penalizing large player coefficients.

The RAPM objective is roughly

fit possession margins + penalize large player coefficients.

That is a useful objective for estimating lineup-adjusted point impact. If we change the response, design matrix, or penalty, the reported metric changes with it. The metric does not capture health, contract value, aging, playoff fit, role portability, or tactical responsibilities that are absent from the design matrix.

A narrower claim supported by the model is: “In this sample, under this lineup-adjusted and regularized model, this player has the higher estimated impact.” If nearby estimates overlap, the comparison should report that as well. Rankings can overstate conclusions in two ways: they can hide uncertainty, and they can hide what quantity was ranked in the first place.

20.3.2 Reporting Rankings

Overstated claim	Model-supported claim
This is the eighth best player.	This player has the eighth highest point estimate, with uncertainty that overlaps several nearby players.
Team A is better than Team B.	The model estimates Team A higher; in the posterior draws, Team A is stronger than Team B 68% of the time.
The metric proves Player X is elite.	The metric is evidence about Player X’s estimated lineup-adjusted point impact in this sample and model. It should be interpreted alongside role and context.
The ranking is objective.	The ranking follows from choices about response, predictors, target, shrinkage, and validation.

Table 20.2: Language for rankings and derived player metrics.

20.4 From Estimates to Claims

The two case studies use different models, but they raise the same reporting problem. A model output is not yet a finished claim. To move from estimate to claim, we need to say what quantity was estimated, what uncertainty attaches to it, and what modeling choices define it. Before presenting a result, the reported claim should answer the questions in Table 20.3.

Question	What to report
Target	What quantity is being estimated: a mean, probability, rank, causal effect, or decision value?
Data	Which seasons, leagues, games, players, or plays were used? What population is the analysis intended to describe?
Variables	What outcome was modeled? What information did the model use? What important information was omitted?
Objective	What did the fitting procedure optimize: squared error, log loss, likelihood, posterior probability, a penalty, or expected utility?
Uncertainty	What changes when we resample, refit, use posterior draws, change priors, or perturb reasonable modeling choices?
Validation	Was the model checked on held-out data, future seasons, calibration, posterior predictive checks, or stability tests?
Decision	If an action is recommended, what objective turns the estimate into that action?

Table 20.3: A short checklist for moving from a model output to a statistical claim.

20.5 Takeaways

- Statistical honesty means matching the claim to the target, data, model, and uncertainty.
- A decision recommendation is conditional on an objective function.
- A small estimated edge should be reported as small, even when it favors one action.
- Rankings should include uncertainty about the order and identify the quantity being ranked.
- A derived metric is defined by its response, predictors, and fitting objective; it is not a direct measurement of total player quality.
- A limitation matters when it changes how the estimate should be used.

References

- [BYW] Brill, R. S., Yurko, R., & Wyner, A. J., *Analytics, Have Some Humility: A Statistical View of Fourth-Down Decision Making*, The American Statistician, 2025.
- [ET] Efron, B., & Tibshirani, R. J., *An Introduction to the Bootstrap*, Chapman and Hall, 1993.
- [BDA] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B., *Bayesian Data Analysis*, CRC Press, 2013.